

CS & IT ENGINEERING



Computer Network

Transport Layer

Lecture No. - 10



By - Abhishek Sir



Recap of Previous Lecture



Topic

TCP Congestion Control



Topics to be Covered



Topic

TCP Congestion Control

V.IMP

Topic

TCP Window Scaling

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
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[NAT]

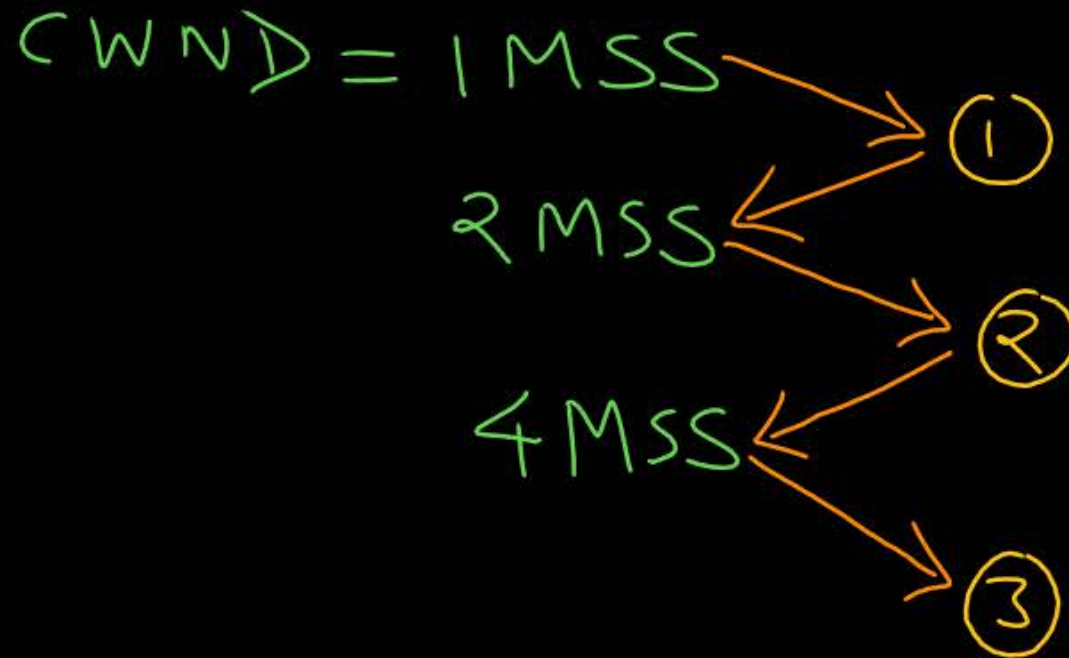
[GATE-2024][2 Mark]



#Q. Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (in MSS) during the third RTT will be _____.

$CWND = 12 \text{ MSS}$
Time-out $\rightarrow *$

$$\begin{aligned} SSThresh &= \frac{CWND}{2} \\ &= \frac{12 \text{ MSS}}{2} = 6 \text{ MSS} \end{aligned}$$



Ans = 4

#Q. Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (in msec) by the TCP connection to get back to 32 KB congestion window is _____.

$$CWND = 32KB$$

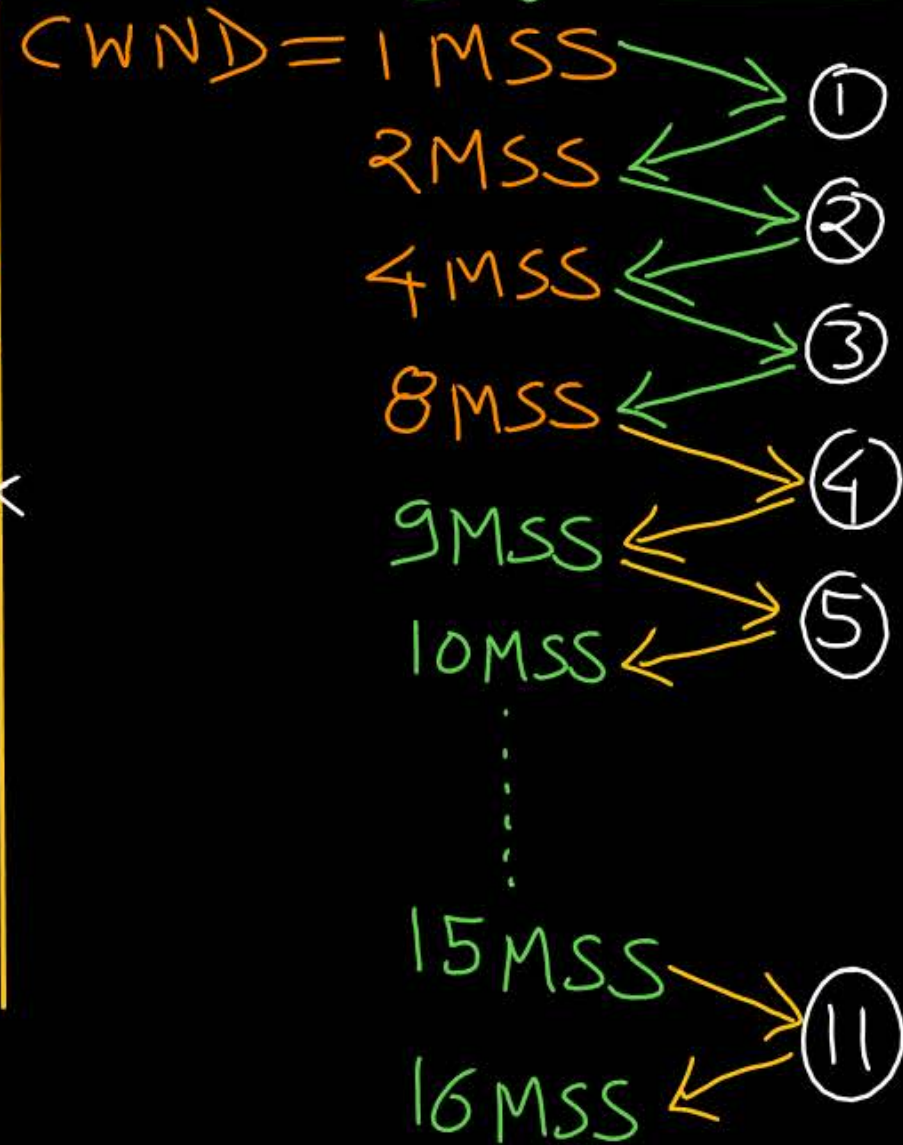
$$= \frac{32KB}{2KB} = 16MSS$$

$$CWND = 16MSS$$

Time-out $\swarrow$ *

$$SSThresh = \frac{CWND}{2}$$

$$= \frac{16MSS}{2} = 8MSS$$



[GATE-2014, Set-1, 2-Mark]

Time taken

$$= 11 * RTT$$

$$= 11 * 100ms = 1100ms$$

$$RTT = 100ms$$

$$MSS = 2KB$$

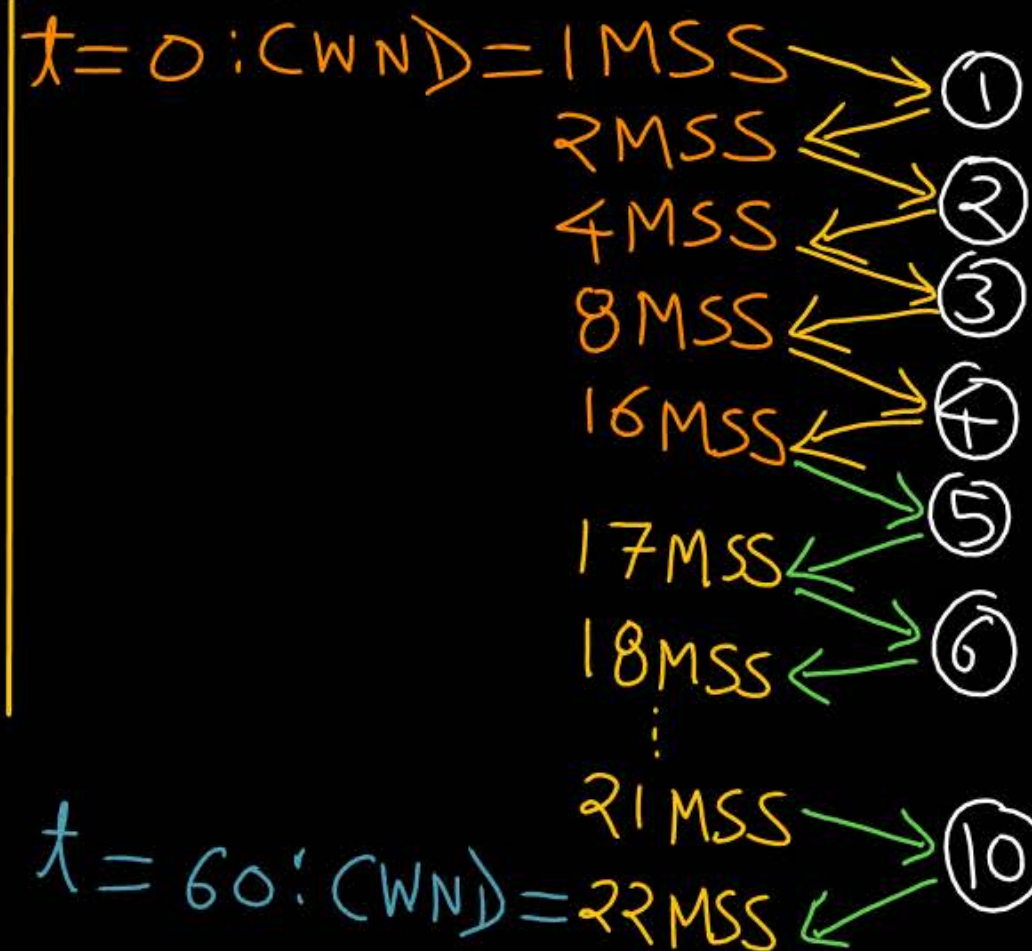
IIT-KGP Ans

$$= 1100 - 1300$$

#Q. Consider a TCP connection between a client and a server with the following specifications; the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time $t=0$. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time $t+60$ ms after all acknowledgments are processed is _____

$$RWND = 50 \text{ KB} \\ = \frac{50 \text{ KB}}{2 \text{ KB}} = 25 \text{ MSS}$$

$$SSThresh = 32 \text{ KB} \\ = \frac{32 \text{ KB}}{2 \text{ KB}} = 16 \text{ MSS}$$



[GATE-2020, 2-Mark]

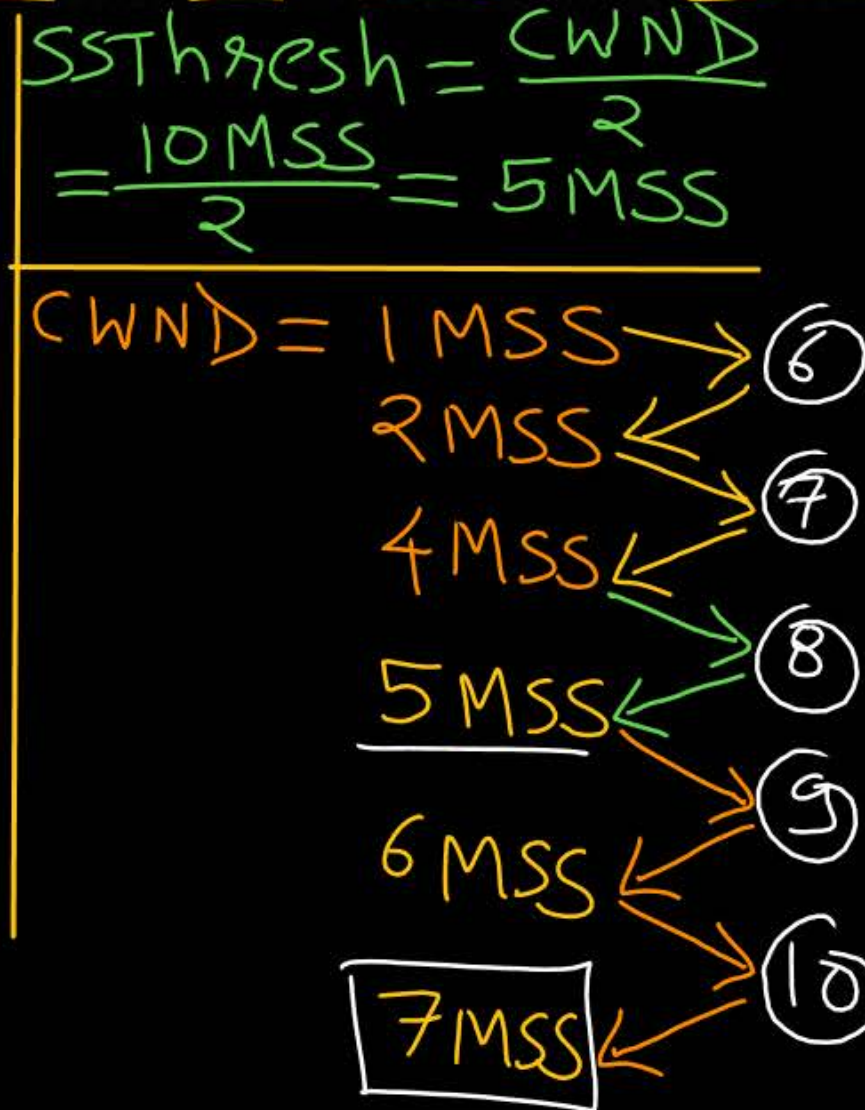
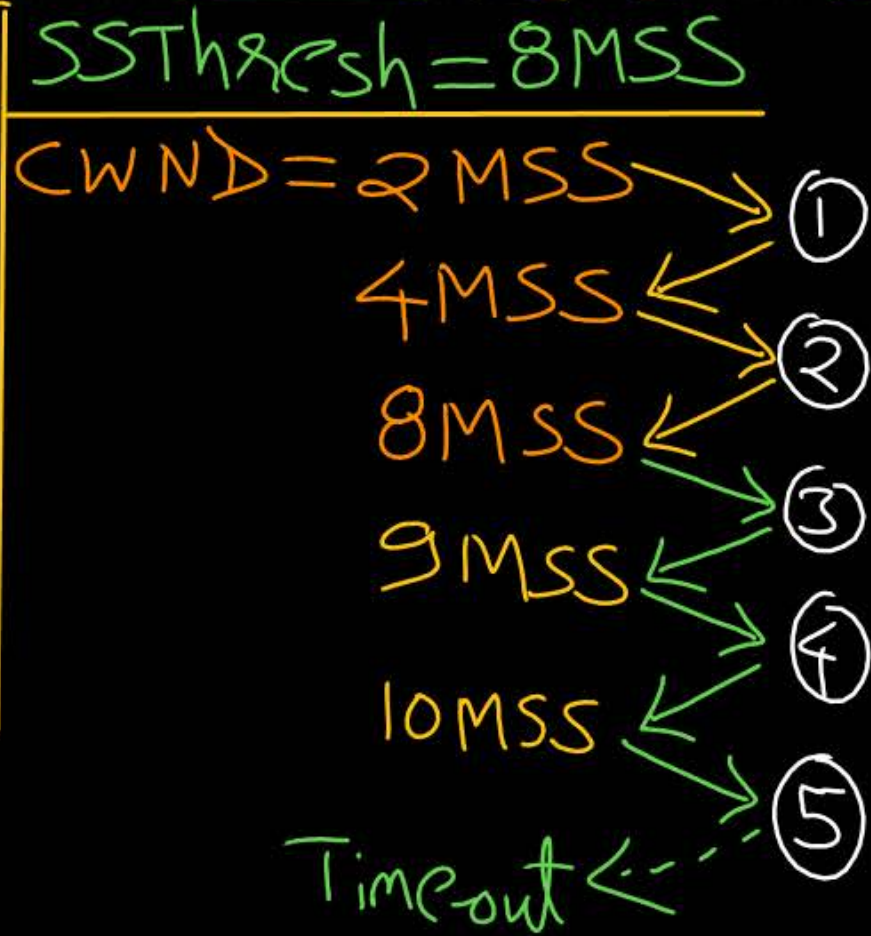
$$RTT = 6 \text{ ms} \\ MSS = 2 \text{ KB}$$

$$\text{Ans} = 22 \text{ MSS} \\ = 22 * 2 \text{ KB} \\ = 44 \text{ KB}$$

$$\boxed{\text{Ans} = 44}$$

#Q. Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

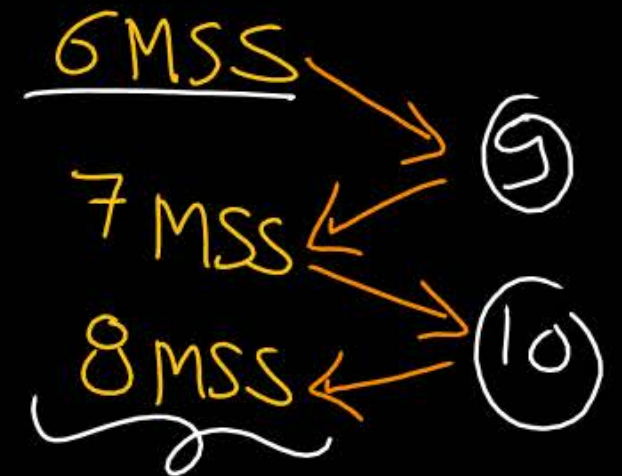
- (A) 8 MSS
- (B) 14 MSS
- ✓ (C) 7 MSS
- (D) 12 MSS



[GATE-2012, 2-Mark]

IIT-D

Marks to ALL



#Q. Suppose that the maximum transmit window size for a TCP connection is 12000 bytes. Each packet consists of 2000 bytes. At some point of time, the connection is in slow-start phase with a current transmit window of 4000 bytes. Subsequently, the transmitter receives two acknowledgments. Assume that no packets are lost and there are no time-outs. What is the maximum possible value of the current transmit window?

- (A) 4000 bytes
- (B) 8000 bytes
- (C) 10000 bytes
- (D) 12000 bytes

[GATE-2004]

IIT-J, H.W.



Topic : Retransmission Policy of TCP

→ Retransmission on time-out
[Retransmit that segment for which time-out occur]

→ Retransmission on receiving 3 duplicate acknowledgments
[Retransmit the unacked segment with smallest sequence number]
[Do not wait for time-out, "Fast Retransmit", Early retransmit]





Topic : TCP

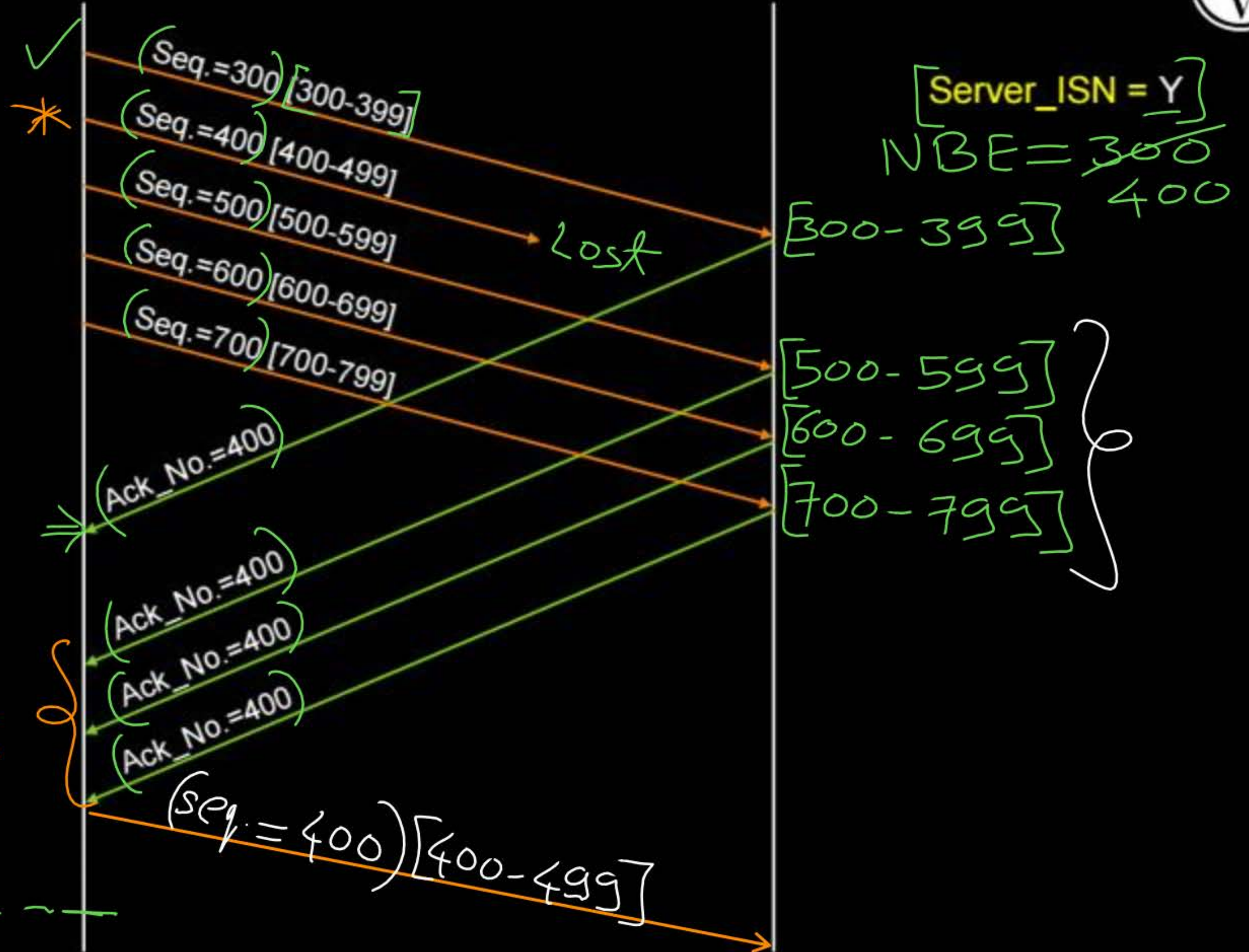


[Client_ISN = 299]

[MSS = 100 bytes]

TCP Client

TCP Server





Topic : Congestion Control

→ In case of "Fast Retransmit" : "Normal Recovery" [As per Time-out]

Update : $SSThresh = [CWND / 2]$

$CWND = 1 MSS$

[* slow start]

→ In case of "Fast Retransmit" : "Fast Recovery"

Update : $SSThresh = [CWND / 2]$

$CWND = SSThresh + 3 MSS$

$\Rightarrow$ Cong. Avoidance

#Q. Which of the following statements are TRUE?

(S1) TCP handles both congestion and flow control TRUE

(S2) UDP handles congestion but not flow control FALSE

(S3) Fast retransmit deals with congestion but not flow control TRUE

(S4) Slow start mechanism deals with both congestion and flow control FALSE

[GATE-2008, 2-Mark]

II SC

(A) S1, S2 and S3 only

✓ (B) S1 and S3 only

(C) S3 and S4 only

(D) S1, S3 and S4 only

Ans: B



Topic : Bandwidth Delay Product

→ bandwidth delay product = $\text{Bandwidth} * \text{RTT}$
[maximum amount of data that can be in transit on a network]

→ Receiver window size field is 16-bits long

→ Maximum possible available space in receiver's receiving buffer
can be advertised without window scale is $(2^{16} - 1)$ bytes

→ if bandwidth delay product is greater than $(2^{16} - 1)$ bytes
than window scaling is required



2 mins Summary



Topic

TCP Congestion Control ✓

Topic

TCP Window Scaling



THANK - YOU

